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Project Report: Design and Implementation of a Digital Clock and Stopwatch

Program: B.S. Electrical Engineering (4th Semester)

1. Abstract

This report details the design, simulation, and hardware implementation of a 24-hour digital clock and a functional stopwatch. The core logic of the system is built entirely using discrete 74-series TTL integrated circuits (ICs) and a 555 Timer. The project demonstrates the practical application of sequential logic design, specifically utilizing cascading asynchronous counters (74LS90) and BCD-to-7-segment decoders (74LS47) to process and display time accurately.

2. Introduction

Digital clocks and stopwatches are fundamental examples of digital system design, relying on precise timing and sequential state changes. Instead of using a microcontroller, this project utilizes raw digital logic. The system is divided into three main stages:

- **Clock Pulse Generation:** Creating a precise 1Hz signal.
- **Counting Logic:** Dividing the frequency into seconds, minutes, and hours using Modulo-N counters.
- **Display Logic:** Decoding binary data into human-readable decimal formats.

The stopwatch functionality builds upon the clock's foundation by introducing control logic (Start, Stop, and Reset) to gate the clock pulse and clear the counter states on demand.

3. Components Required

- **NE555 Timer IC:** For the 1Hz Astable Multivibrator.
- **74LS90 (Decade Counter):** Used as Mod-10 and Mod-6 counters.
- **74LS47 (BCD to 7-Segment Decoder):** Translates 4-bit BCD to display signals.
- **7-Segment Displays:** Common Anode type (compatible with 74LS47 active-low outputs).
- **Logic Gates:** 74LS08 (AND), 74LS32 (OR), 74LS04 (NOT) for resetting logic and control switching.
- **Passive Components:** Assorted resistors (e.g., 330Ω for displays), capacitors, and push buttons.
- **Power Supply:** 5V DC regulated.

4. Circuit Theory and Design

4.1. Clock Generation (The 555 Timer)

To keep accurate time, the counters require a 1Hz clock pulse. The 555 timer is configured in astable mode to generate a continuous square wave. The frequency is determined by the resistor and capacitor values using the standard formula:

$f = 1.44 / ((R1 + 2R2) \times C)$

For a 1Hz frequency (1 pulse per second), specific values like R1 = 1 kΩ, R2 = 72 kΩ, and C = 10 μF are selected and tuned.

4.2. Counting Logic (Seconds and Minutes)

Both the seconds and minutes sections operate on a Modulo-60 base. Each requires two 74LS90 ICs:

Units Digit (Mod-10): The 74LS90 naturally counts from 0 to 9. The clock pulse feeds into CPA, and QA is tied to CPB to enable standard BCD counting. When it resets from 9 back to 0, its highest bit (QD) provides a falling edge to trigger the next stage.
Tens Digit (Mod-6): Time does not count to 99; it rolls over at 59. Therefore, the second 74LS90 must reset when it hits 6 (binary 0110). We achieve this by wiring outputs QB and QC into the Master Reset pins R0(1) and R0(2).

4.3. Counting Logic (Hours)

For a 24-hour clock, the system must reset to 00 when it reaches 24.
The units digit acts as a Mod-10 counter, and the tens digit as a Mod-3 counter.
An AND gate monitors the outputs representing "2" on the tens counter and "4" on the units counter.
When both are high, the output of the AND gate triggers the Master Reset of both hour counters.

4.4. Stopwatch Control Logic

To transition the circuit from a continuous clock to a stopwatch, additional control elements are introduced:
Start/Stop: An AND gate is placed between the 555 Timer output and the first counter's clock input. A toggle switch or flip-flop circuit controls the other leg of the AND gate, effectively allowing or blocking the 1Hz pulse.
Reset: A push-button is connected to the Master Reset pins of all 74LS90 ICs in the stopwatch circuit, allowing the user to clear the displays to zero asynchronously.

5. Simulation and Implementation

- Before hardware assembly, the logic was rigorously simulated using Proteus. This allowed for the verification of the Mod-6 and Mod-24 reset conditions without risking hardware damage.
- The 74LS90 pin configurations and cascading triggers were monitored using Proteus's logic probes.
- Once verified, the schematic was transitioned to a breadboard. Careful attention was paid to grounding and supply decoupling to prevent false clocking from switch bounce or power supply noise.

6. Challenges and Troubleshooting

- **Switch Bounce:** Initially, the mechanical push buttons for the stopwatch controls caused multiple counts per press. This was mitigated by adding debouncing capacitors across the switches.
- **Floating Inputs:** TTL logic (like the 74-series) interprets floating inputs as logic HIGH. All unused inputs, particularly reset pins on the 74LS90, had to be explicitly tied to ground to ensure stable operation.

7. Conclusion

The successful implementation of the digital clock and stopwatch validated fundamental digital logic concepts. Utilizing standard discrete ICs like the 74LS90 provided deep, hands-on insight into state machines, asynchronous cascading, and BCD decoding. This project successfully bridged the gap between theoretical digital systems and practical hardware troubleshooting.